

Data Management

Date: 2026-03-24

Overview

The application stores its operational data in the Django database.

Database modes in the current build:

- PostgreSQL when `DB_HOST` is configured
- SQLite when `DB_HOST` is not configured

Main Stored Entities

The core stored entities in the current data model are:

- `User`: Django authentication user
- `UserProfile`: application role attached to a user
- `Room`: collaborative room
- `RoomMember`: membership link between user and room
- `Post`: chat message in a room
- `Activity`: activity definition, phases, and per-activity agent settings
- `Agent`: automated intervention definition
- `Intervention`: agent-generated message
- `EvidenceNudgeState`: state for evidence-related nudges
- `SessionSummary`: generated session summary stored as JSON sections
- `FinalAnswerSelection`: chosen final answer for a run
- `FinalAnswerVote`: vote for a final answer

Data Relationships

High-level relationships:

- a Django `User` has one `UserProfile`
- a room can have many members, posts, interventions, and summaries
- a room can reference a selected activity
- an activity stores phases as JSON
- summaries are linked to both a room and an `activity_run_id`

Summary Data Storage

Session summaries are stored in `SessionSummary` with four main JSON fields:

- `participation_data`
- `process_data`
- `quality_data`
- `extracted_content`

These fields store derived summary information rather than a separate full copy of raw chat history.

Current summary generation includes items such as:

- participation per member
- phase-level activity statistics
- intervention counts by rule
- quality flags such as lack of evidence or unbalanced participation
- extracted decisions, action items, unanswered questions, and outcomes

PDF Output

Summary data can be exported to PDF using ReportLab. The PDF is an export format derived from a stored `SessionSummary`; the summary record itself remains in the database.

Authentication And Role Data

Role handling is stored in `UserProfile.role`.

Important implementation note:

- the current model code exposes `learner` and `facilitator` as the active role choices
- Django admin access remains available separately through Django authentication

Profile records are created automatically when a Django user is created.

Seed And Fixture Data

Seeded activities are stored in:

- `backend/message_board/fixtures/activities.json`

Load them with:

```
cd backend
```

```
python manage.py loaddata message_board/fixtures/activities.json
```

Inventory Deliverable

The coursework software and data inventory is delivered separately as:

- `software_data_inventory_final.xlsx`

Retention And Backup Considerations

No automated backup or retention system is included in the prototype.

If the project is taken forward, the maintainer should define:

- how long room and summary data should be retained
- whether temporary learner and facilitator accounts should be cleaned up periodically
- how backups and restores will be performed for the chosen database

Data Management Limitations

- no formal retention policy is implemented
- no export capability beyond PDF summaries is included
- no backup scripts are included
- no audit log of admin actions is included